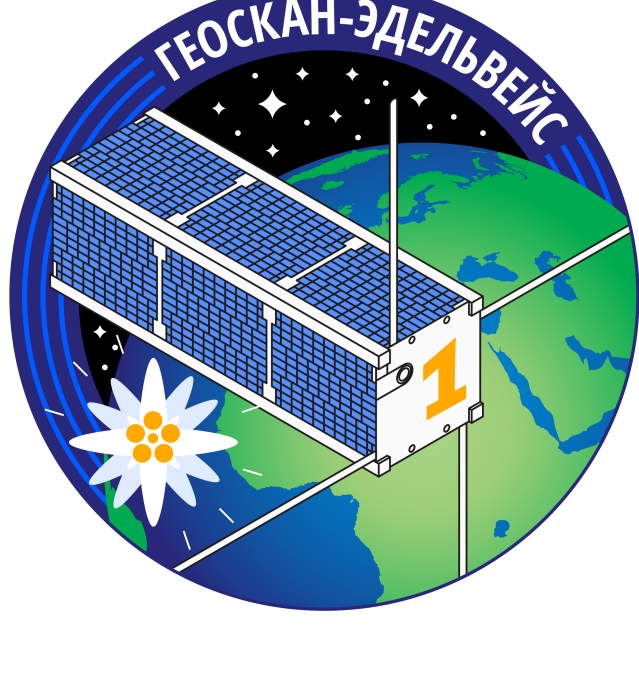




Geoscan-Edelveis radio protocol

1. Protocol version

2.0

2. Data transmission parameters

UHF band parameters:

- downlink frequency – 436.200 MHz
- baudrate – 9600
- signal modulation type – GFSK
- CRC16-CCITT
- PN9 Scrambling
- Fixed packet size – 64 bytes (without CRC)
- little-endian

3. Beacon packet structure in AX.25 format

Table 1: AX.25 structure

Size, bytes	Description	Value	Offset, bytes
6	Destination Callsign	0x84 0x8A 0x82 0x86 0x9E 0x9C (BEACON)	0
1	Destination SSID	0x60	6
6	Source Callsign	0xA4 0xA6 0x64 0x60 0xA6 0x40 (RS20S)	7
1	Source SSID	0xE1	13
1	Control field (UI)	0x03	14
1	Protocol ID (without Layer 3 implementation)	0xF0	15
26	Telemetry		16

Table 2: Telemetry values

Description	UoM	Type
Time	unix format	uint32
Current consumption	A (*0.0000766)	uint16
Current from the panels	A (*0.00003076)	uint16
Battery 1 voltage	V (*0.00006928)	uint16
Total battery voltage	V (*0.00013856)	uint16
Temperature on the X+ panel	°C	uint8
Temperature on the X- panel	°C	uint8
Temperature on the Y+ panel	°C	uint8
Temperature on the Y- panel	°C	uint8
Temperature on the Z+ panel	none	uint8
Temperature on the Z- panel	°C	uint8
Battery 1 temperature	°C	uint8
Battery 2 temperature	°C	uint8
CPU load	% (*0.390625)	uint8
Total reboots at OBC	(value minus 7476) pcs.	uint16
Total reboots at CommU	(value minus 1505) pcs.	uint16
RSSI	(value minus 1505) pcs.	uint8

4. Radio packet structure

The threshold for determining the sync byte should be set to 16 bits.

Table 3: Radio packet structure with CRC16-CCITT checksum

Size, bytes	Purpose	Description	Offset	Value
4	Preamble	Modem bit synchronization	0	0xAAAAAAAA
4	Sync word		4	0x930B51DE
2	Header	Data packet service information	8	
62	Data	Data	10	
2	CRC	CRC-16 checksum	72	

To calculate CRC16, 0x1021 polynomial and 0x0000 start value are used.

Table 4: Structure of the radio packet header

Byte number	Description	
1	Satellite number	01
2	Reserved	00

Table 5: Image transmission data structure

When a file is being transmitted by the satellite, each 64-byte packet has the following data field structure

Size, bytes	Description	Offset, bytes
2	Header	0
1	Data field size	2
2	Message type	3
2	Packet offset	5
1	Subsystem number	7
56	Payload	8

Example of a data packet during image transmission

[01 00] [3E] [05 09] [9C 0B] [0A] [69 6E 33 A2 B7 5B 6B DB 64 B9 88 6E 46 51 B1 4F 02 3F 61 F8 D6 64 8F 84 65 70 CB 22 F0 F9 E3 06 9D 68 27 BD 55 96 39 D6 DA 58 BE 4C 2A F0 E3 B1 FC EA 9D D5 D5 E3 DD 3C]

When a large file is being sent, the data is transmitted by the satellite sequentially, packet by packet, 56 bytes of payload at a time.

The following software can be used to receive photos from Geoscan-Edelveis satellite:

SA2KNG geoscan-tools : <https://github.com/kng/geoscan-tools>

UBIQBJ Geoscan-Edelveis-Decoder : <https://github.com/Foxiks/Geoscan-Edelveis-Decoder>